ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** force / area	Formula.
2	**(c)** eye	Calm centre.
3	**(d)** air pressure	Atmospheric pressure.
4	**(c)** pascal	Pa.
5	**(b)** wind speed	Wind.
6	**(c)** Atlantic	Region.
7	**(c)** increase pressure	Small area.
8	**(a)** rotating funnel	Whirlwind.
9	**(c)** seriously	Danger.
10	**(d)** reduce pressure on sand	Wide feet.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force acting per unit area (2).	2
12	Pressure due to weight of air (2).	2
13	Wind from sea to land (1); Blows during the day (1).	1+1
14	P = F / A (1); Pascal (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	P = F/A (1); 40 / 2 (1); 20 Pa (1).	1+1+1
16	Stay indoors (1); Listen to radio (1); Move to shelters (1).	1+1+1
17	Uproots trees (1); Damages houses (1); Cuts power lines (1).	1+1+1
	OR: P = F/A (1); 90 / 3 (1); 30 Pa (1).	1+1+1
18	Sun heats the land unevenly (1); Warm air rises (1); Cool air rushes in (1).	1+1+1
	OR: Cyclone is large, tornado small (1); Cyclone over the sea (1); Tornado is over land (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	P = F/A (1); Pressure rises when area falls (1); Knife (1); Camel (1); Tractor (1).	1+1+1+1+1
	OR: Weight of air (1); Suction cup (1); Straw (1); Syringe (1); Barometer (1).	1+1+1+1+1
20	200/4 = 50 Pa (1); 200/0.5 = 400 Pa (1); Smaller area, larger pressure (1); Units pascal (1); Eight times (1).	1+1+1+1+1
	OR: Charge in clouds (1); Lightning (1); Thunder (1); Stay indoors (1); Avoid trees (1).	1+1+1+1+1

Mistakes to avoid:

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** warm sea	Warm water.
2	**(d)** bottom	Pressure.
3	**(c)** weight	Air exerts pressure.
4	**(b)** day	Land warm.
5	**(d)** atmospheric pressure	Air pushes liquid.
6	**(a)** moving air	Air in motion.
7	**(d)** night	Land cools.
8	**(d)** doubles	P is proportional to F.
9	**(c)** rises	Lighter.
10	**(b)** lightning	Same discharge.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Barometer (2).	2
12	Wind from land to sea (1); Blows at night (1).	1+1
13	Small area gives high pressure (1); Enters wood easily (1).	1+1
14	Calm centre (1); Low pressure region (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Needle (1); Knife (1); Nail (1).	1+1+1
16	Pressure changes with height (1); Ear pressure differs (1); Swallowing helps (1).	1+1+1
17	Steam is trapped (1); Pressure rises (1); Boiling point increases (1).	1+1+1
	OR: Camel's feet (1); Wide tyres (1); School bag strap (1).	1+1+1
18	Farming (1); Fishing (1); Travel (1).	1+1+1
	OR: Suction cup (1); Straw (1); Syringe (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Make three holes (1); Fill water (1); Observe streams (1); Lowest goes farthest (1); Pressure depends on depth (1).	1+1+1+1+1
	OR: Pressure rises with depth (1); Wall must withstand pressure (1); Thick base (1); Saves collapse (1); Example (1).	1+1+1+1+1
20	Uneven heating (1); Warm air rises (1); Cool air fills (1); Sea breeze day (1); Land breeze night (1).	1+1+1+1+1
	OR: Funnel (1); Rotates fast (1); Touches ground (1); Destroys buildings (1); Warning needed (1).	1+1+1+1+1

Mistakes to avoid:

- Writing P = A / F or forgetting that the area must be in square metres (convert cm squared first).
- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** halves	P is inversely proportional to A.
2	**(a)** atmospheric pressure	Barometer.
3	**(b)** stay indoors	Safe.
4	**(b)** high to low pressure	Pressure difference.
5	**(b)** depth	More liquid above.
6	**(b)** wide tyres	Reduces pressure.
7	**(a)** thunderstorm	Storm.
8	**(d)** low pressure centre	Low pressure.
9	**(b)** height	Less air.
10	**(a)** storm and flood	Safety.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	High water pressure at depth (1); Suit protects the body (1).	1+1
12	Air moving horizontally (1); Due to pressure difference (1).	1+1
13	Cyclone (1); Tornado (1).	1+1
14	Stay indoors (1); Avoid water and trees (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Increases with depth (1); Acts in all directions (1); Same level, same pressure (1).	1+1+1
16	Funnel of air (1); Spins fast (1); Very destructive (1).	1+1+1
17	Torch (1); Dry food and water (1); First aid kit (1).	1+1+1
	OR: Strong winds (1); Heavy rain (1); Storm surge (1).	1+1+1
18	Direction (1); Time of day (1); Temperature of land and sea (1).	1+1+1
	OR: Anemometer: wind speed (1); Barometer: pressure (1); To forecast weather (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Warm ocean (1); Air rises (1); Low pressure (1); Air spirals in (1); Eye forms (1).	1+1+1+1+1
	OR: Farmers (1); Fishermen (1); Pilots (1); Government (1); Public (1).	1+1+1+1+1
20	Before: keep supplies (1); Before: secure house (1); During: stay in shelter (1); After: avoid wires (1); Boil water (1).	1+1+1+1+1
	OR: Direction (1); Time (1); Temperature of land (1); Pressure (1); Name (1).	1+1+1+1+1

Mistakes to avoid:

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- Dividing the weight by the area of one foot when the person stands on both feet.
- Saying liquid pressure depends on the shape or width of the container; it depends on depth.
- Saying wind blows from low to high pressure, or that the eye of a cyclone is the stormiest part.
- Writing that atmospheric pressure increases with height.
- Confusing pressure (N/m squared, pascal) with force (N, newton).

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